

ASSIGNMENT 1

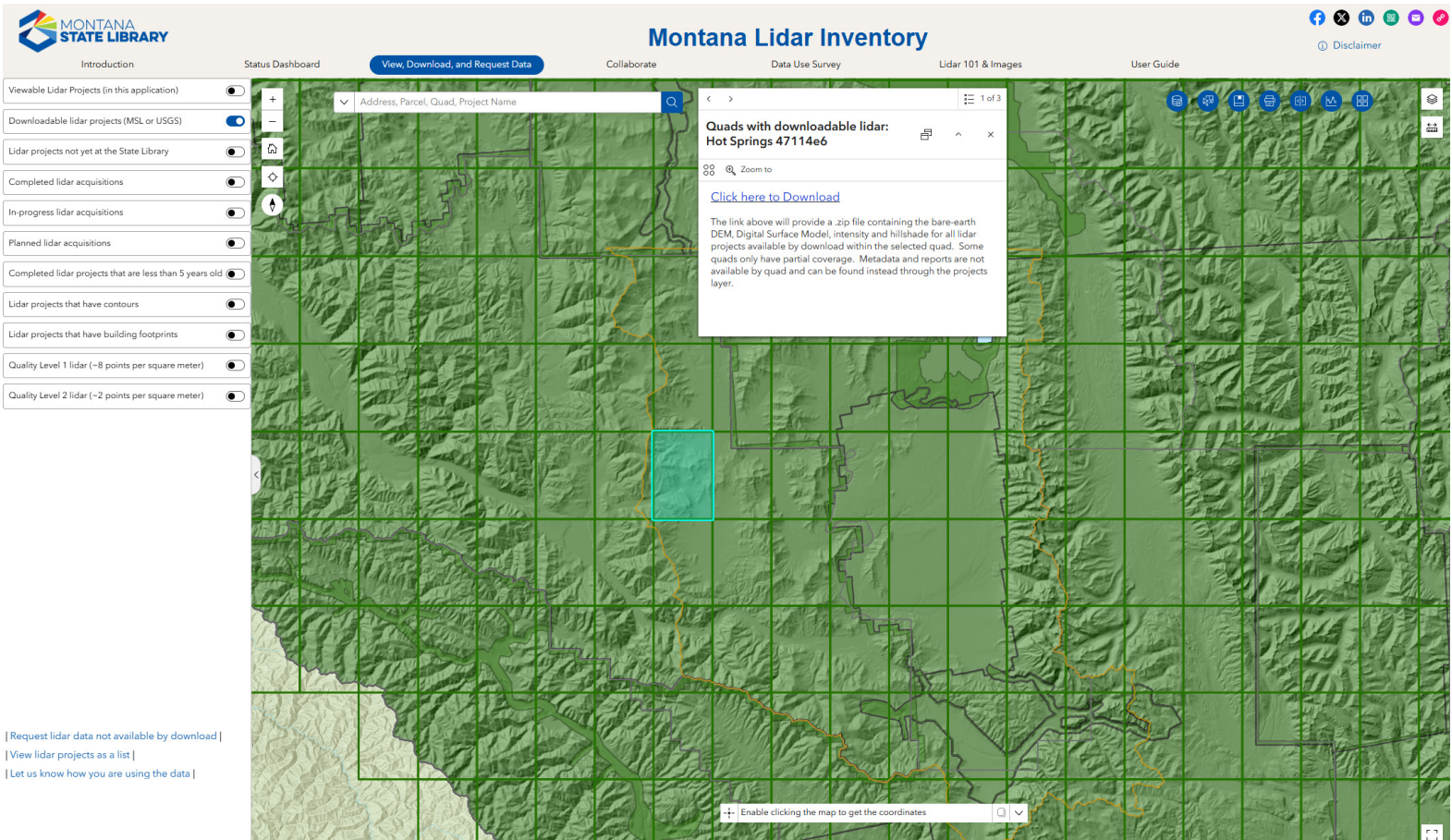
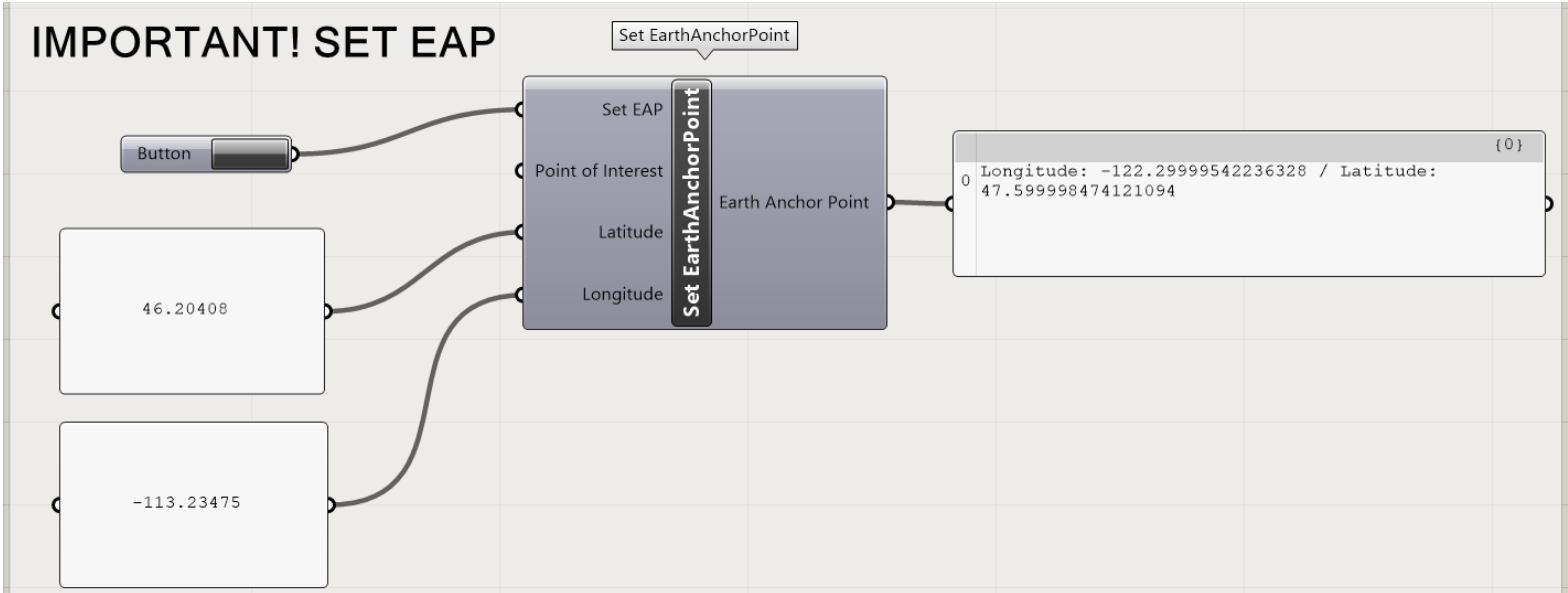
LiDAR TOPOGRAGHY DATA & LAZER CUTTING

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ARCH 565 - Advanced Computer Applications

For my site, I selected my family cabin near Georgetown Lake.

- Although most of Montana has LiDAR data available, there was no coverage for this rural area. Because of that, I could not use the typical LiDAR and Cloud Compute workflow and had to find another method.
- I went to the Montana State Library website and downloaded a ZIP file for the “Quad” that includes my cabin site.
- After unzipping the folder, I located a DEM file, which I imported into Rhino using the Heron plugin.
- I installed the Heron plugin through Rhino’s Package Manager.
- In Grasshopper, the first step was to set the Earth Anchor Point. This establishes the base reference point so all imported geographic data aligns correctly in the model.



Topography + Contours

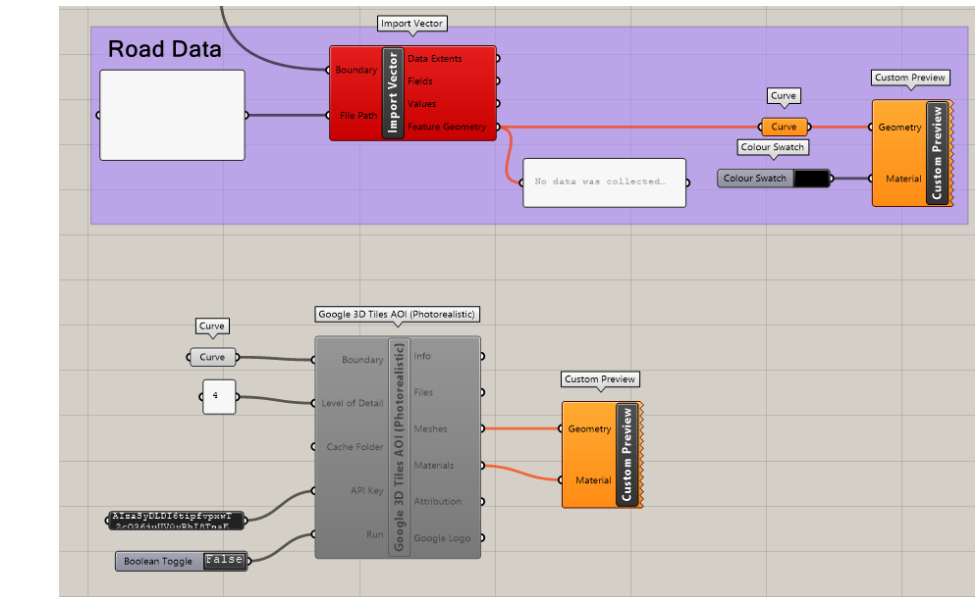
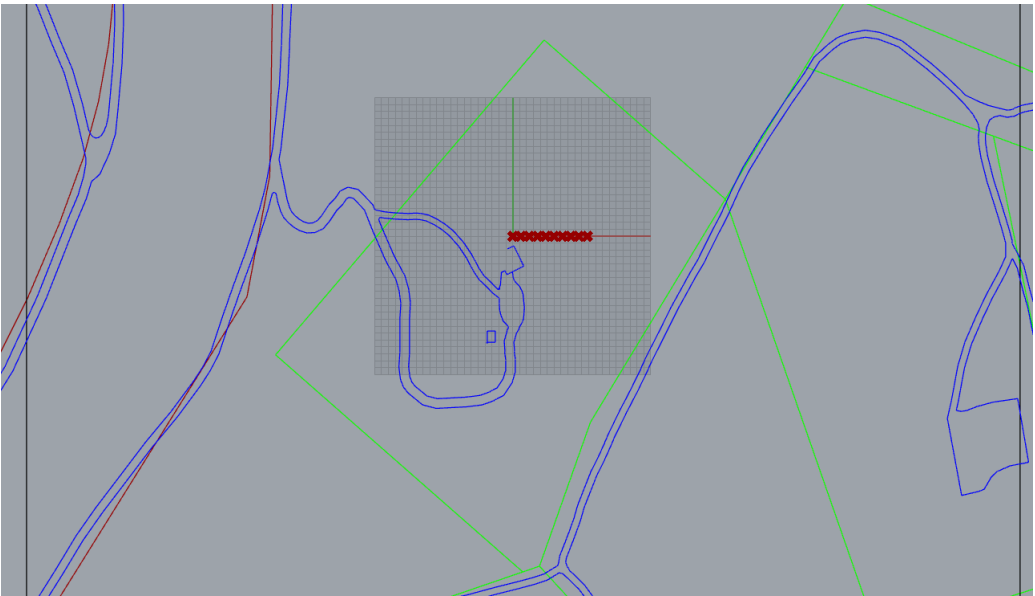
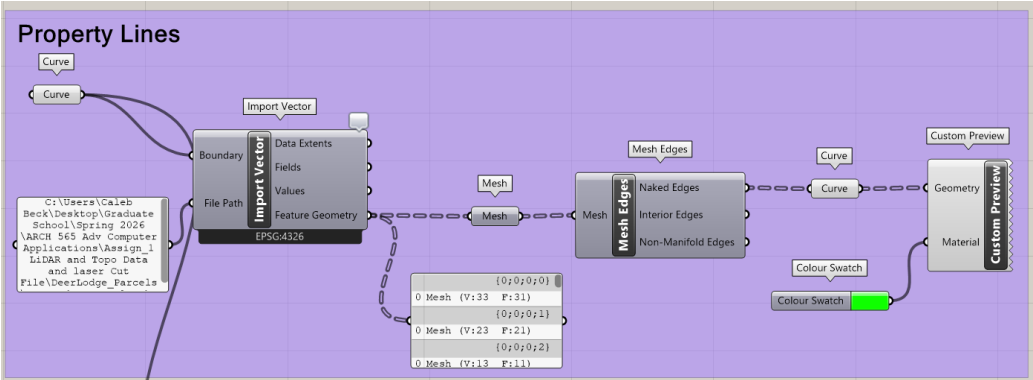
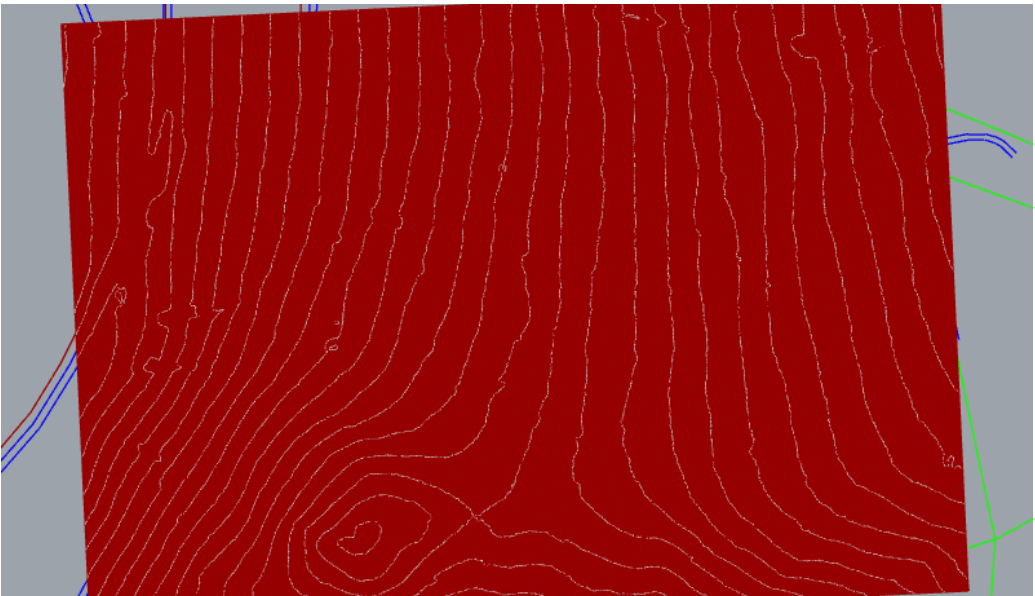
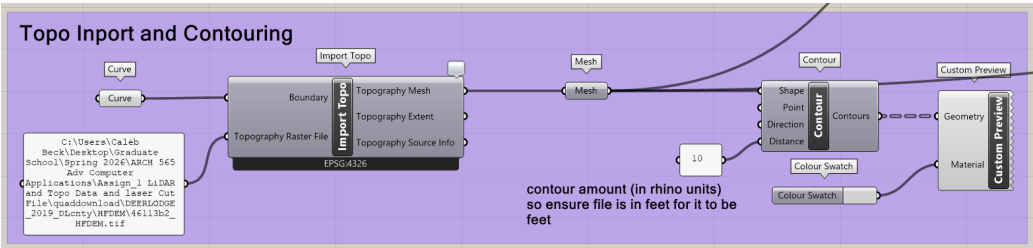
- Using Heron, I imported the topo data and generated a mesh from the DEM.
- I then created contour lines from the mesh at 10-foot intervals.
- The topo mesh can be previewed in Grasshopper using the Custom Preview component.

GIS Data (Property Lines)

- I also imported GIS data for property lines.
- Through Montana Cadastral, I downloaded parcel data for Deer Lodge County, where the cabin is located.
- I linked the parcel file to the Grasshopper script using a file path panel.

OSM Data (Roads) & Google 3D Tiles

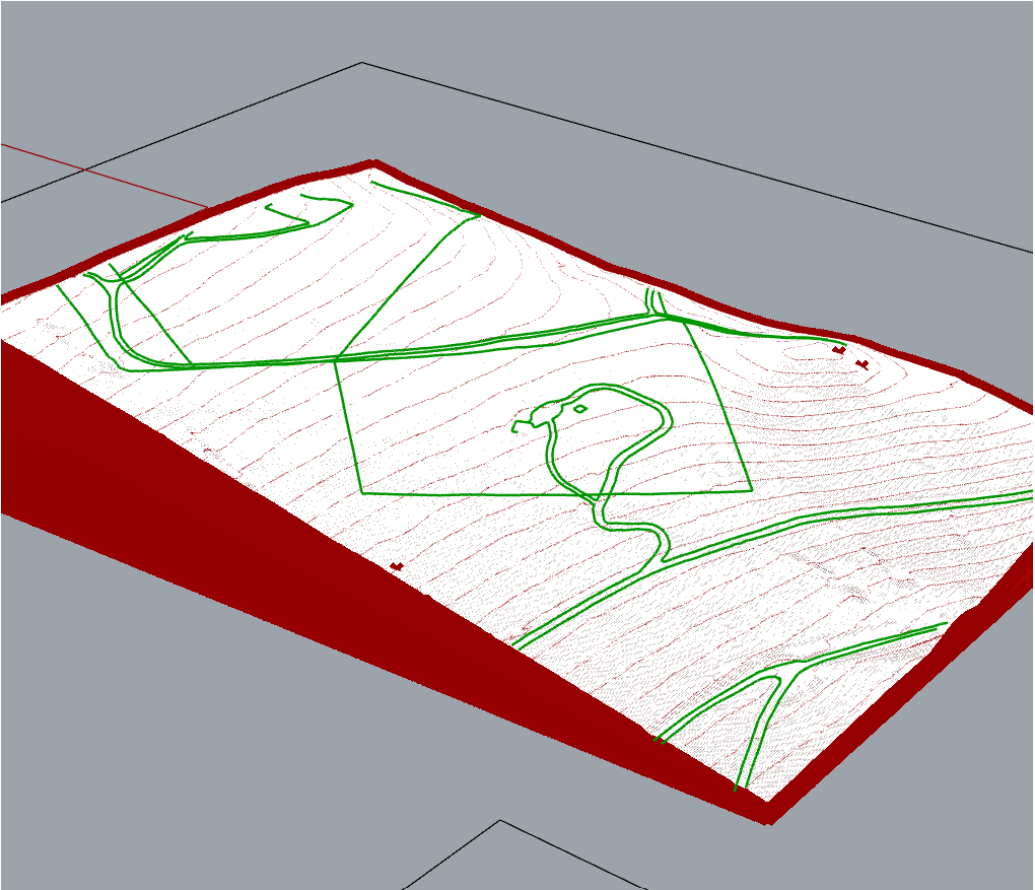
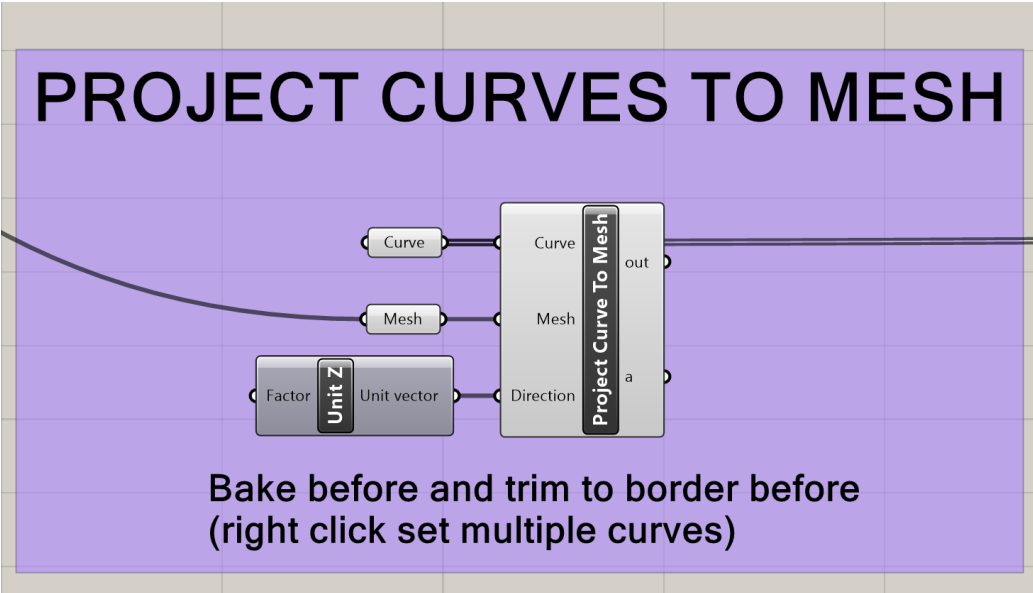
- I attempted to import OSM road data.
- I tried OpenStreetMap & Google 3D Tiles, but there was no available data for this area.
- Because of this, I manually traced the roads using Google Earth.



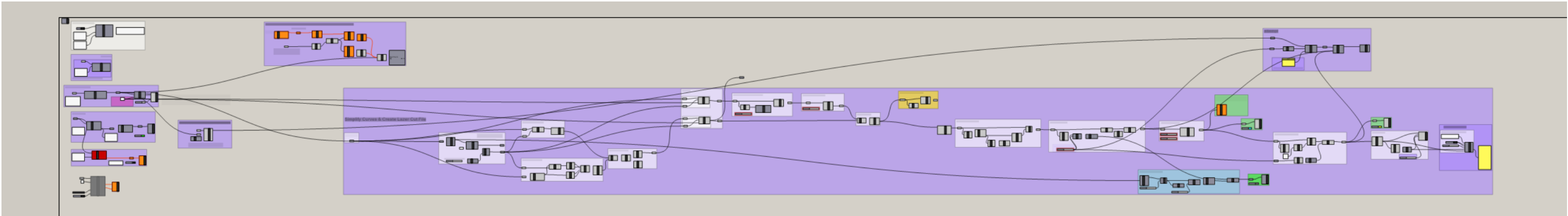
No Data

- Now that we have all of our vector data, the next step is to project those curves onto the topography mesh.
- With Chris’s help and his GitHub code prompt, I created a Custom Project Curve to Mesh node.
 - After creating the node, I baked the curves into Rhino and trimmed them to the site boundary.
 - I then set up the inputs: connected the topo mesh, placed all curves into a curve container, and defined the Z-axis as the projection direction.
 - Once the curves are projected onto the mesh, the digital model is complete. It can now be baked and further developed in Rhino or in Revit using Rhino.Inside.

```
Grasshopper - Project curve to Mesh  
  
A = Rhino.Geometry.Curve.ProjectToMesh(Curve, Mesh, Vector, Tolerance)
```

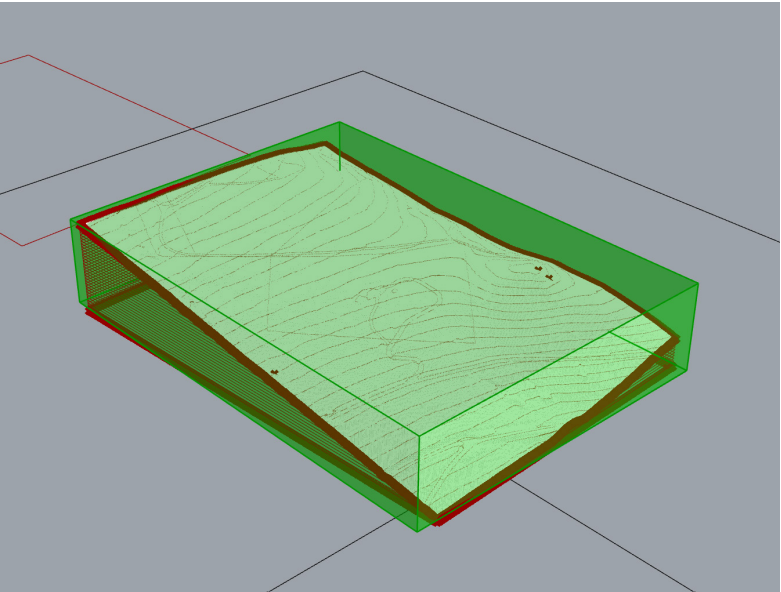
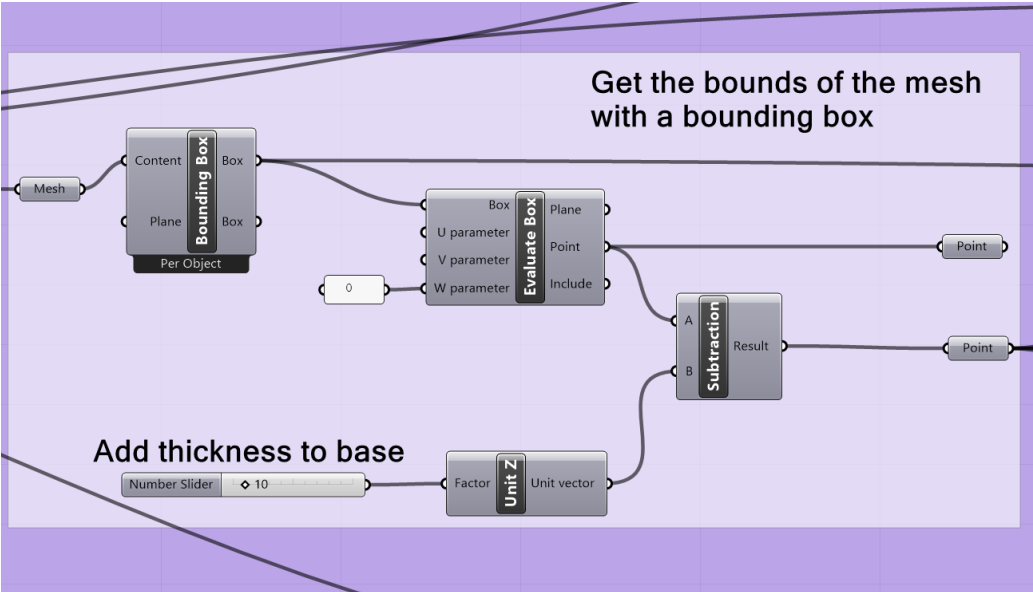


Now that the digital model is complete, the remaining work will concentrate on digital fabrication. It is worth noting that nearly half of the script shown below is dedicated to refining curves and geometric inconsistencies. While this improves fabrication efficiency and reduces processing time, it is not strictly necessary for the assignment. Accordingly, I will focus specifically on the portions of the script that directly address the fabrication workflow.



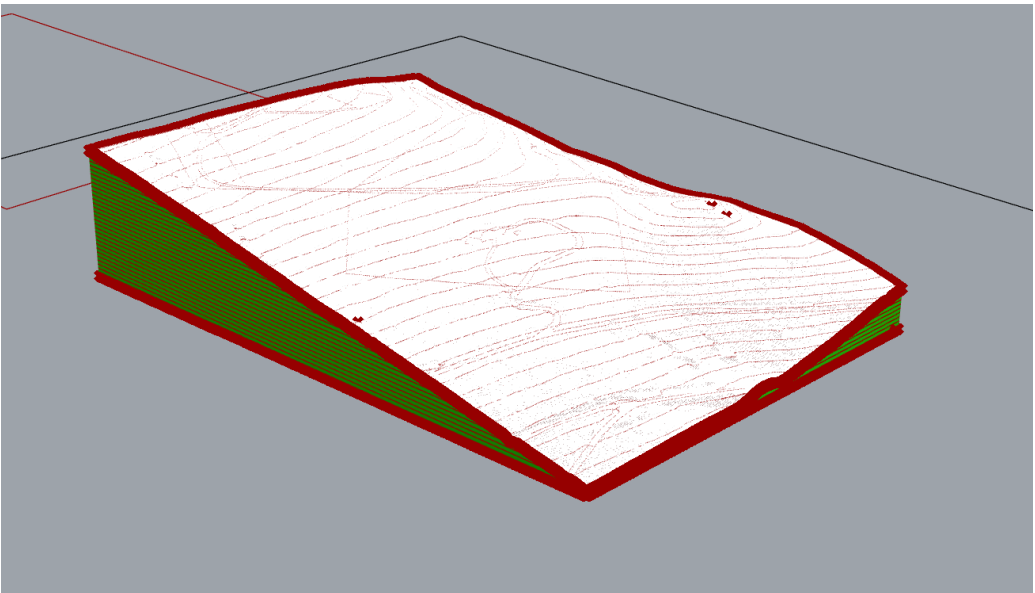
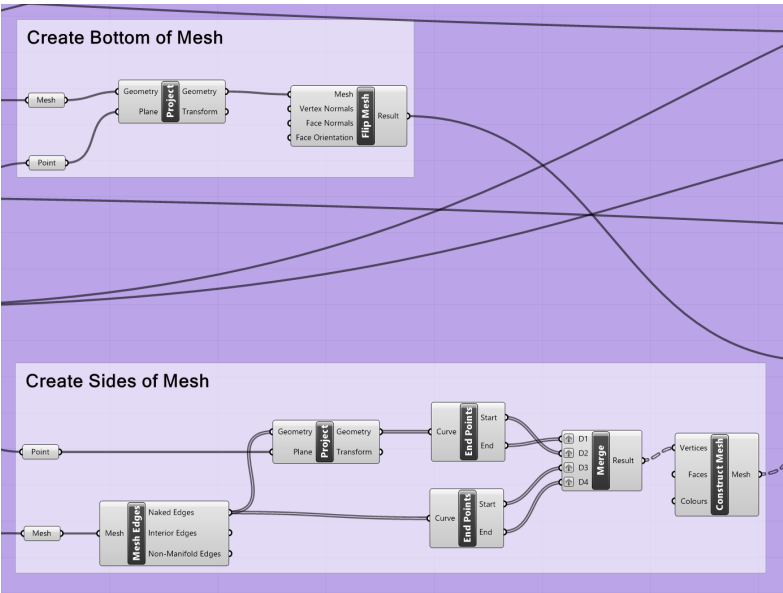
Bounding Box

- First step, the mesh edges need to be cleaned up. This ensures clean 90° corners.
- A bounding box is created and everything outside the box is trimmed away. This gives the model a clean rectangular base.
- The box can be edited to make the base thicker.



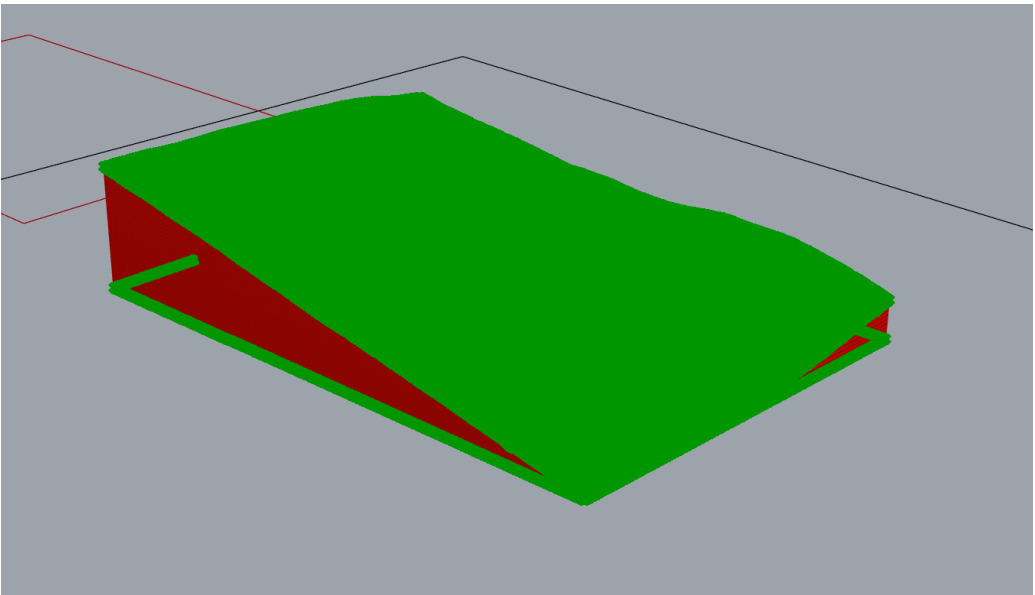
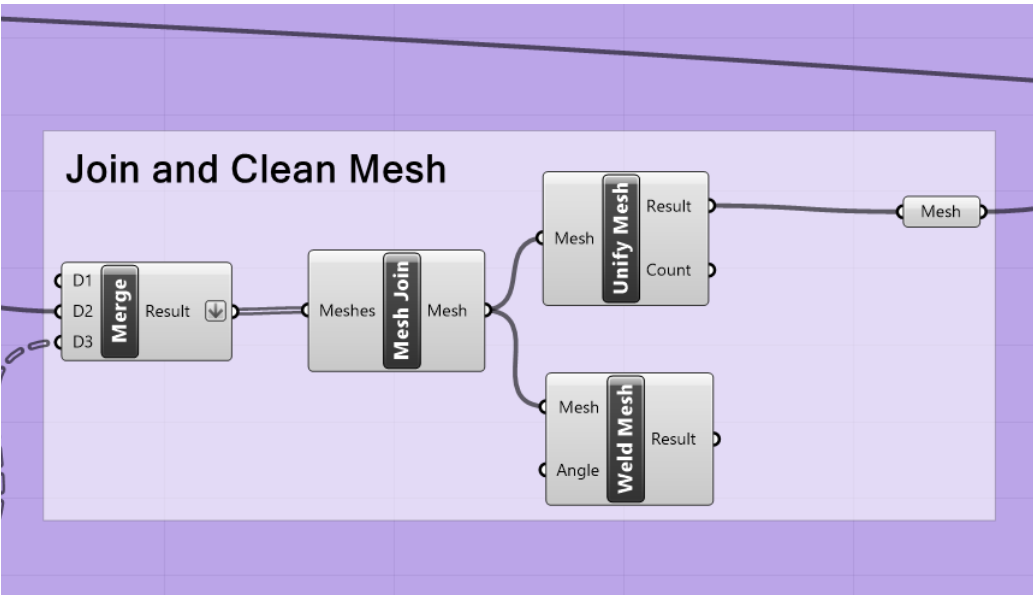
Forming the Base and Walls

- Using the trimmed mesh, the next step is to create the bottom and side faces.
- The “Project” nodes are used to generate the bottom surface, and then a mesh is built between the top and bottom to form the sides.
- This will be crucial for the contouring step.



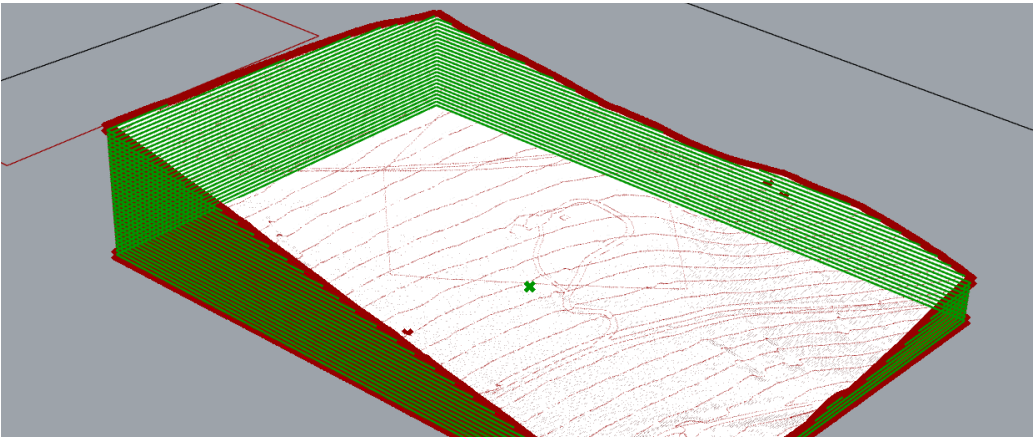
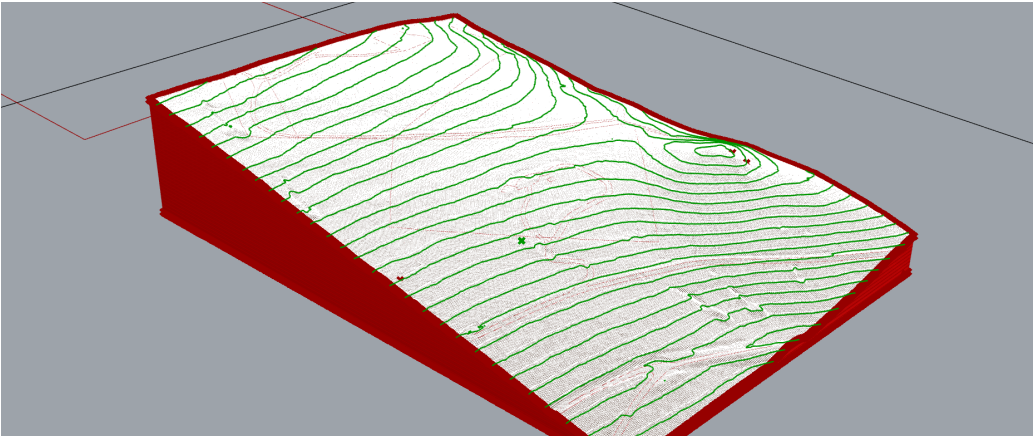
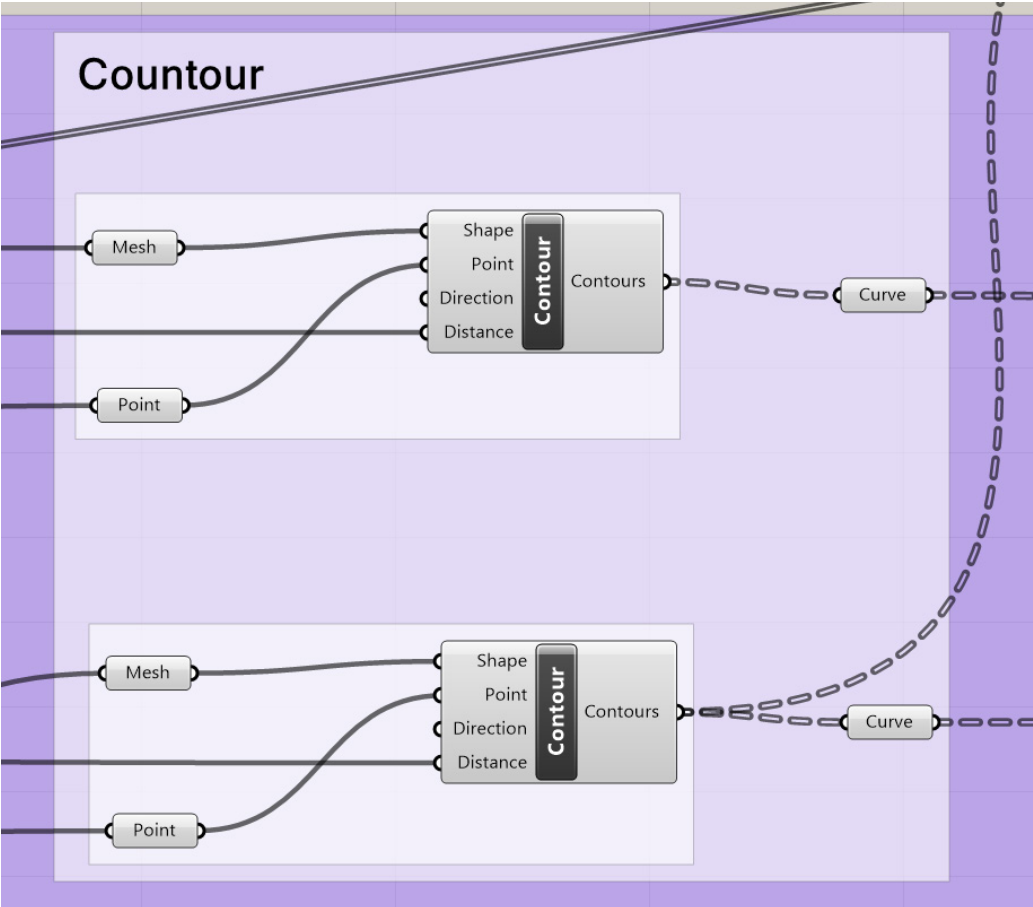
Joining and Cleaning Mesh

- Now that we have all the sides to our mesh, we can use the “mesh join” node and connect all faces.
- We can also clean up these connections with nodes such as “Weld Mesh & Unify Mesh”
- The mesh is ready to contour.



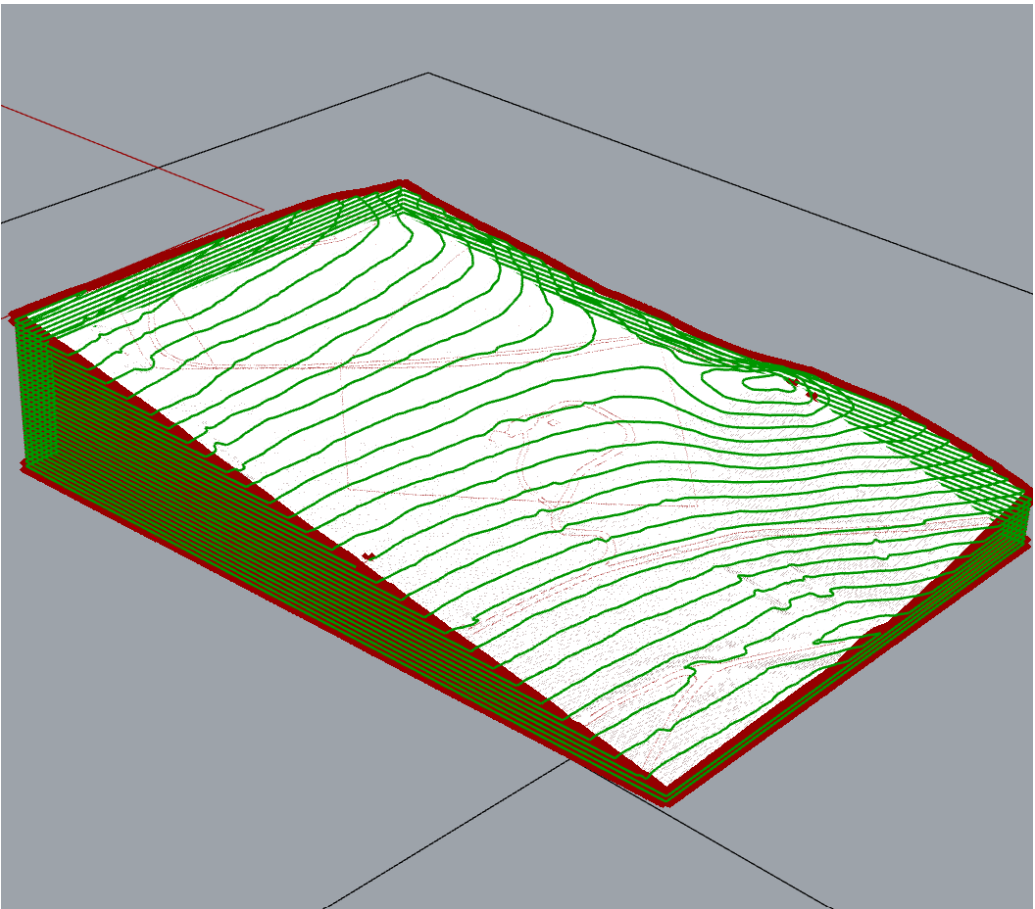
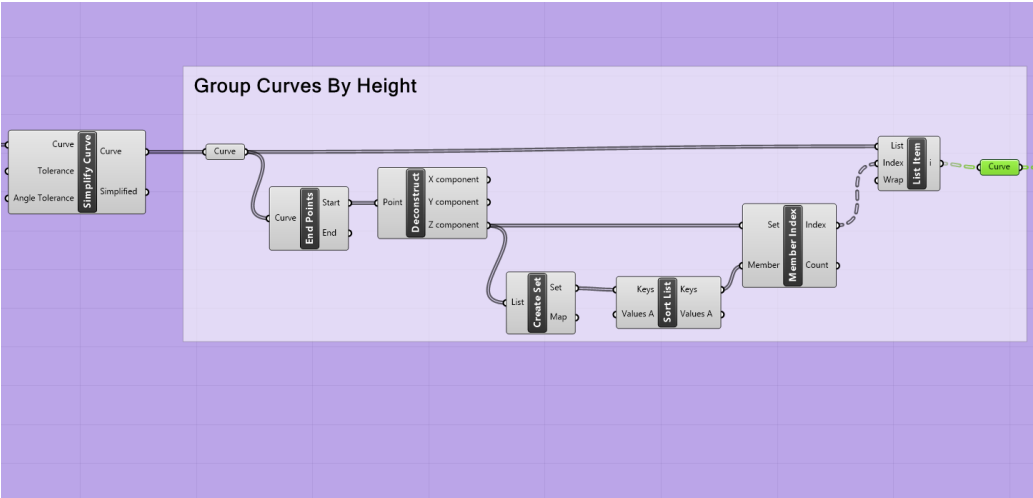
Contouring the Mesh

- Split the mesh into two contour nodes: one for the top and one for the sides.
- Connect each mesh to its contour node and set the base point.
- You can use the contour numbers from the digital model to match the physical model, or set them separately if needed.
- Ensure the top and side contours use the same number for accuracy.



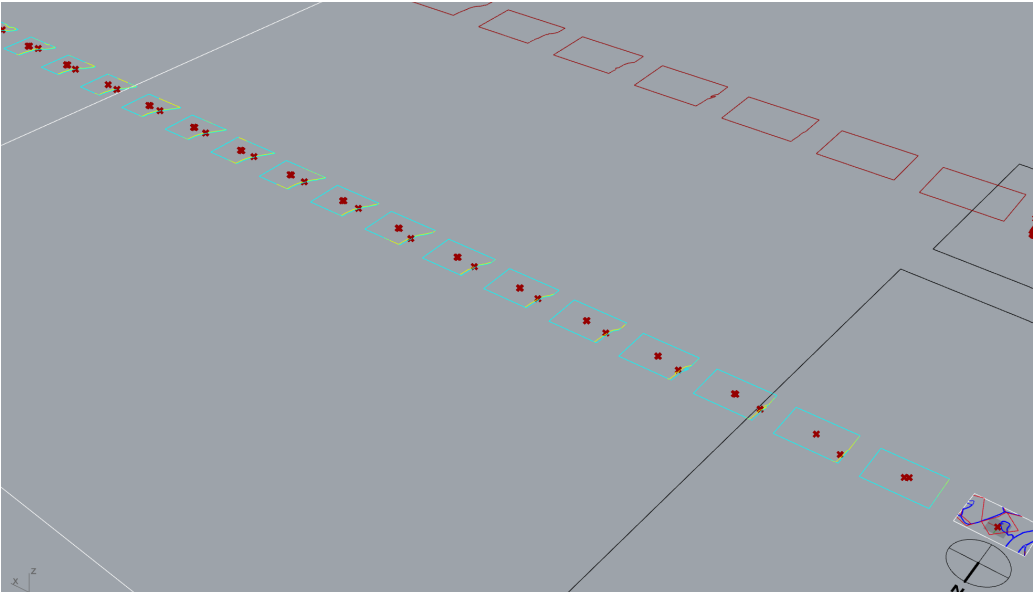
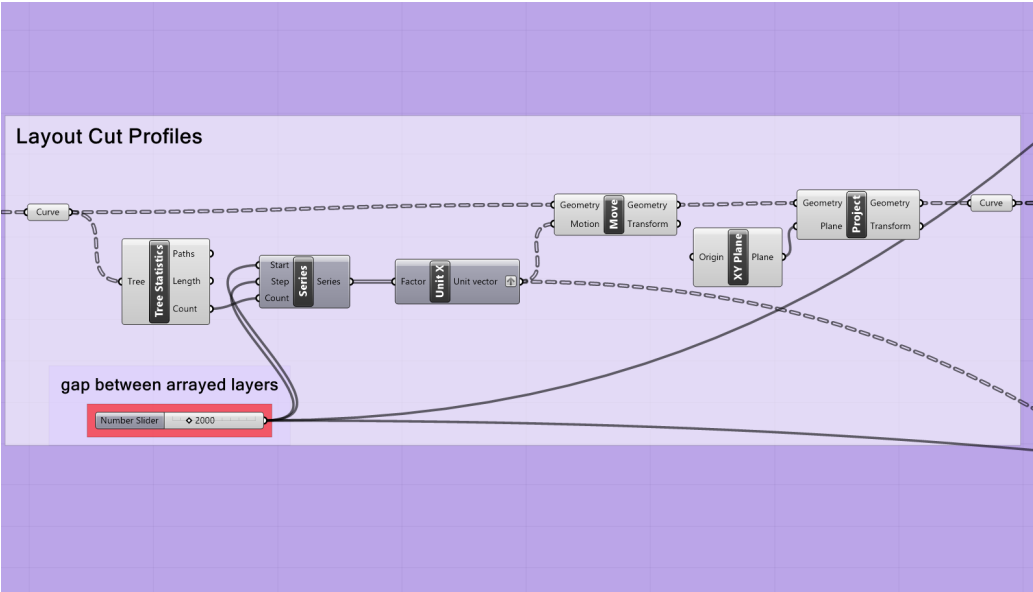
Grouping Contours

- Now that you have both contours, we can join them.
- First, we simplify the curves, and deconstruct thier endpoints.
- Then we can list those points and index them based off of thier Z axis.
- The outcome is fully closed loops of contours, that we can begin to sepperate and array.



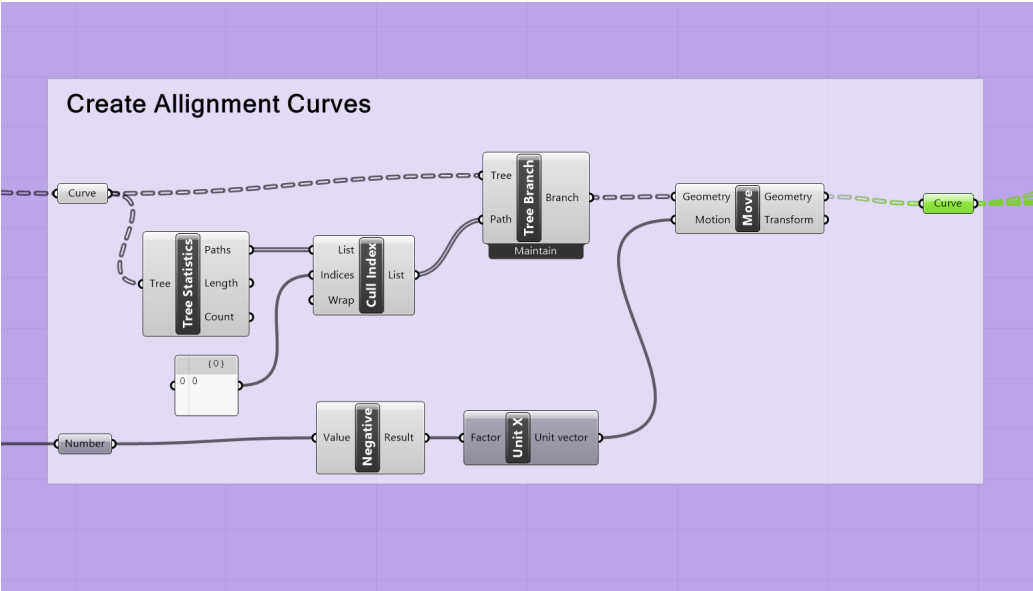
Cut Profile Layout

- To separate the layers into cut-ready planes, we must use series and trees to gather only the necessary data.
- From there, we can project along the X-axis and set the desired distance between the layers.



Alignment Curves

- Using tree data and the Cull Index, we can project the curves of the next layer onto the current cut profile.
- This ensures proper alignment when assembling the model.



Layer Labels

- Finally, to maintain the correct order when assembling, we can add a small number label to each layer.
- Using a Polygon Center node, we locate the center point of each layer.
- Then, we create a 3D text tag and use tree statistics and series to label the layers accurately.

